TYLER COX

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EDUCATION

The Ohio State University Expected May 2027
Bachelor of Science in Computer Science & Engineering
Coursework: Data Structures & Algorithms, Operating Systems, Database Design & SQL, Object-Oriented Programming

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C, Swift
Frameworks: React, Next.js, FastAPI, SwiftUI; REST APIs
Data: PostgreSQL, SQLAlchemy, Prisma, Alembic
Tools: Git, Docker, GitHub Actions, pytest, Vitest, Vercel

PROJECTS

IncidentOPS 2026 - Present
React, TypeScript, Python, FastAPI, PostgreSQL, SQLAlchemy, Docker | Code | Demo
- Built a React/TypeScript incident workbench backed by FastAPI and PostgreSQL, with searchable queues, analyst assignments, investigation notes, and audit histories.
- Enforced Admin, Analyst, and Viewer permissions in the API; implemented Argon2 password hashing, hashed server-side sessions, and CSRF/origin checks.
- Committed incident changes and audit events in one database transaction; added Alembic migrations and validated CSV login-event analysis.
- Automated frontend and backend checks in GitHub Actions CI, including PostgreSQL integration, container restart persistence, and backup/restore tests.

MacroMesh / Calorie Compass 2026 - Present
Next.js, TypeScript, SwiftUI, PostgreSQL, Prisma, OpenAI API | Code | Demo
- Built Next.js web and SwiftUI iOS clients for conversational meal logging, review-before-save, meal history, and progress tracking.
- Integrated a verified food catalog, USDA FoodData Central, Open Food Facts, and LLM fallback with source labels and structured validation.
- Improved branded-food matching from 3/25 to 21/25 passing cases in a fixed offline benchmark; added identity/category regression checks.
- Rejected meal saves that reused a user's idempotency key; used Prisma transactions and database uniqueness constraints, with tests for duplicate-key and conflict handling.

CPU Scheduler Visualizer 2025
JavaScript, React, Vite, Vitest, Testing Library | Code | Demo
- Implemented FCFS, non-preemptive shortest-job-first, priority scheduling, and Round Robin; calculated waiting, turnaround, and response times.
- Built editable workloads and Gantt-chart playback in React; verified scheduling logic and interface behavior with 7 passing Vitest tests.

WORK EXPERIENCE

Resident Advisor | The Ohio State University Housing Aug 2023 - Present
- Respond to resident concerns and on-call incidents, de-escalate conflicts, document reports, and coordinate with housing and campus staff.
- Organize community events, explain housing policies and resources, and prioritize time-sensitive issues alongside routine responsibilities.

Office Assistant | The Ohio State University Housing Sep 2022 - Present
- Triage student and family requests, maintain office records, and coordinate front-desk communications using Excel, Outlook, and Workday.

ADDITIONAL TRAINING

Cisco Python Essentials 2; IBM Cloud Computing Fundamentals; Google Cloud Introduction to Generative AI.